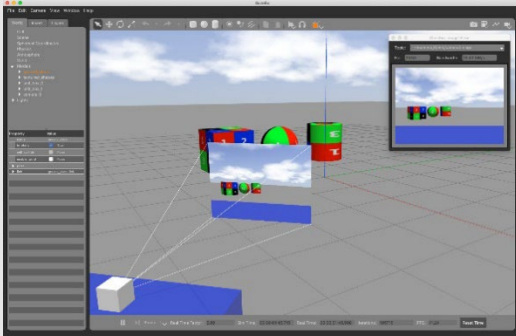


Learning to Remote Autonomous Control of a Mobile Robot using an External Security Camera



Subject:

Interconnected autonomous heterogeneous robots have the potential to surpass the capabilities of individual systems by leveraging the principles of cooperation and machine learning. A mobile robot equipped with onboard sensors, such as laser scanners, often suffers from a limited, local, and low-dimensional view of its environment for navigation around obstacles, especially in the presence of occlusions or a narrow field of view. In contrast, an external security camera can provide a broader and more comprehensive perspective of the environment, capturing global spatial relationships that are not easily accessible to the robot itself. This project explores how to bridge these complementary modalities—onboard mobility and external perception—through interactive safe robot learning, enabling more effective and intelligent remote autonomous control strategies for a mobile robot using an external security camera. You will develop reinforcement learning [1] and/or free-energy minimization [2] algorithms to safely, autonomously, and remotely guide a mobile robot using an onboard laser scanner and an external camera to any given destination in the camera image. You will design autonomous exploration and obstacle mapping methods in the external camera coordinate frame [3, 4] for safe path planning and remote path-following control. Your algorithms will be developed and tested in the Gazebo simulator with Robot Operating System (ROS), Python, and PyTorch. If time allows, you will test them on real TurtleBot3 robots with Bosch security cameras in the ME Robotics Lab.

Assignment:

The objective of this project is to design, implement, and test interactive safe robot learning algorithms (including, but not limited to, reinforcement learning and free-energy minimization) for autonomously and remotely controlling a mobile robot using an external security camera. In addition to contributing to the development of the simulation and experimental setup, you will be expected to deliver a well-documented and experimentally validated software package that can:

- Perform calibration-free robot learning for remote visual servoing of a mobile robot toward a given destination in the external camera image.
- Perform robot learning for safe remote control of a mobile robot around obstacles using external security cameras and an onboard laser scanner.
- Perform autonomous exploration and mapping in the external camera image plane for safe path planning and remote path-following control.

References:

- [1] Brunke et. al., "Safe learning in robotics: From learning-based control to safe reinforcement learning," *Annual Review of Control, Robotics, and Autonomous Systems*, vol.5, no.1, pp.411-44, 2022
- [2] de Vries et.al. "Expected Free Energy-based Planning as Variational Inference," *arXiv preprint, arXiv:2504.14898*, 2025.
- [3] Elfes, "Using occupancy grids for mobile robot perception and navigation," *Computer*, vol. 22, no. 6, pp. 46-57, 1989.
- [4] Juliá et. al., "A comparison of path planning strategies for autonomous exploration and mapping of unknown environments," *Autonomous Robots*, vol. 33, pp. 427–444, 2012.

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